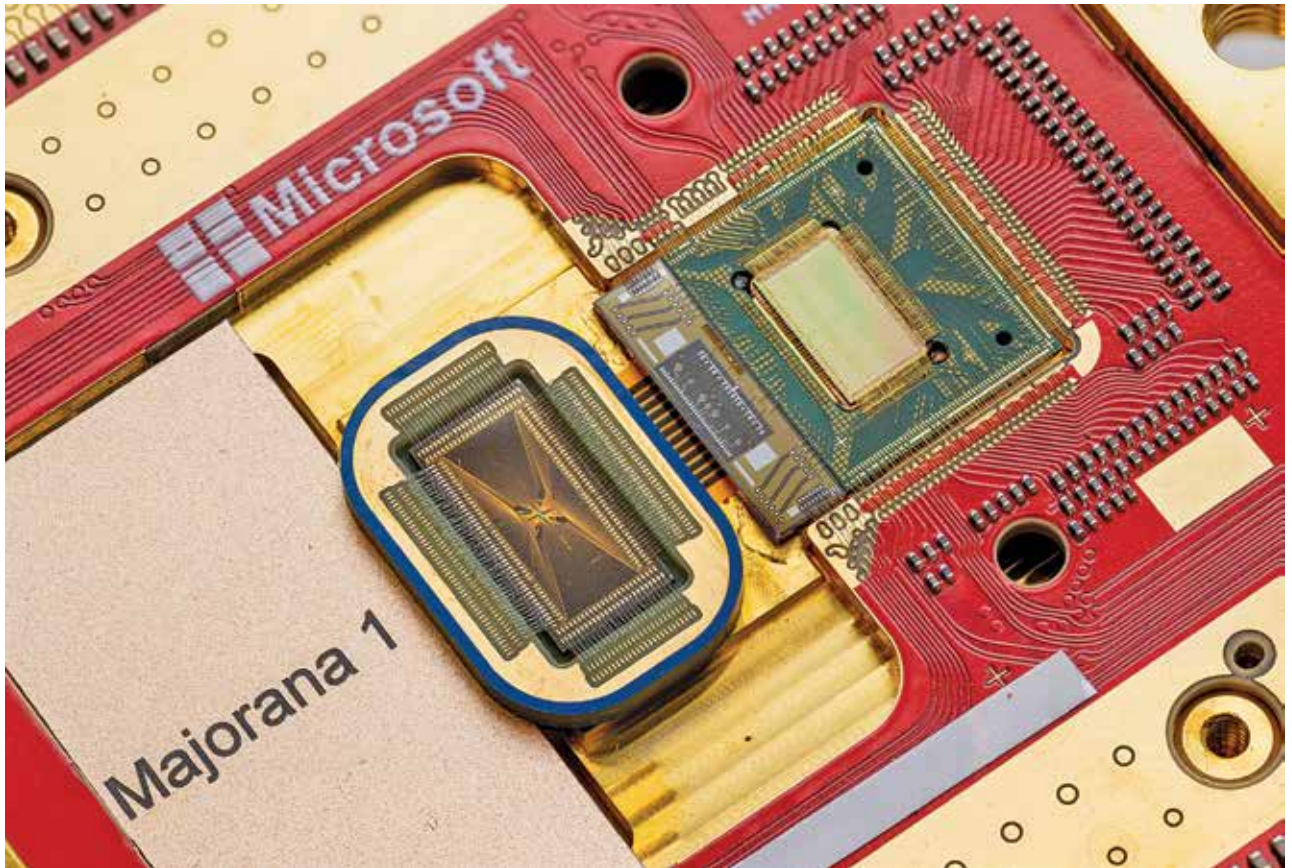


Palm-Sized Quantum Chips Spark Boundless Dreams



Microsoft's Majorana 1 (Courtesy: Microsoft)

Meet the newest entrants to the quantum computing race—Google's Willow, Microsoft's Majorana 1, and Amazon's Ocelot, which take us closer to the possibility of a commercially-viable large quantum computer.

Those interested in quantum computing know that there is constant activity in the space—research and development, government policies, investment, and more. However, when there is a wave of announcements from big tech companies, everyone notices—even the non-techies. The unveiling of quantum chips by Google, Microsoft, and Amazon, in quick succession, has garnered a lot of media traction, suddenly making quantum computing a coffee table topic. This buzz has made

people more aware of what quantum computing can do—and awareness has the potential to accelerate a technology's journey to the mainstream.

Why are companies so keen on developing fault-tolerant large quantum computers? Is it because they can outperform classical computers? Yes, they can. But more than that, quantum computers have massive parallel processing ability far beyond the reach of classical supercomputers because their basic building block—a qubit—can hold multiple states, unlike